STAC WARS · PS2

Solar Radiation: Analysis and Prediction

Voyage-123 observation data, September – December 2016

1. Summary

This report explores the Voyage-123 solar log, explains what drives radiation, builds a day-wise solar report with Pandas/NumPy, and sets out a regression model for prediction. Main findings:

- Radiation follows a bell-shaped daily curve: zero overnight, rising after ~06:00, peaking between about 11:00 and 13:00 (solar noon ≈ 12:20), gone again by ~18:30.
- **Temperature** has the strongest relationship with radiation (r = 0.73). Humidity (–0.23), wind direction (–0.23), pressure (0.12) and wind speed (0.07) are weak.
- The all-time maximum, **1601.26** on **2016-09-04 at 12:15:04**, is about 50% above the clear-sky ceiling (~1050–1100) seen on every other day. It is most likely a cloud-edge spike or sensor artefact, not a typical clear-sky value.
- The five days with the highest mean radiation are all in September, and all are smooth, cloud-free curves.
- Clear-sky peaks fall steadily through the season (≈1070 in Sep → ≈900 in mid-Nov) and the day shortens.

2. Data and method

Columns: Radiation, Temperature, Pressure, Humidity, WindDirection(Degrees), Speed, TimeSunRise, TimeSunSet plus date/time stamps (≈5-minute cadence). Units are not given in the problem statement; the plots are consistent with W/m² for radiation, °F for temperature, inHg for pressure and mph for wind speed, but this is an assumption. The *Data* and *Time* columns were combined into a single Datetime column, the data sorted chronologically, and sunrise/sunset strings converted to minutes so that “minutes after sunrise” could be computed. All code is in src/.

3. Part A & B — Visualisation and inferences

3.1 Radiation versus time of day

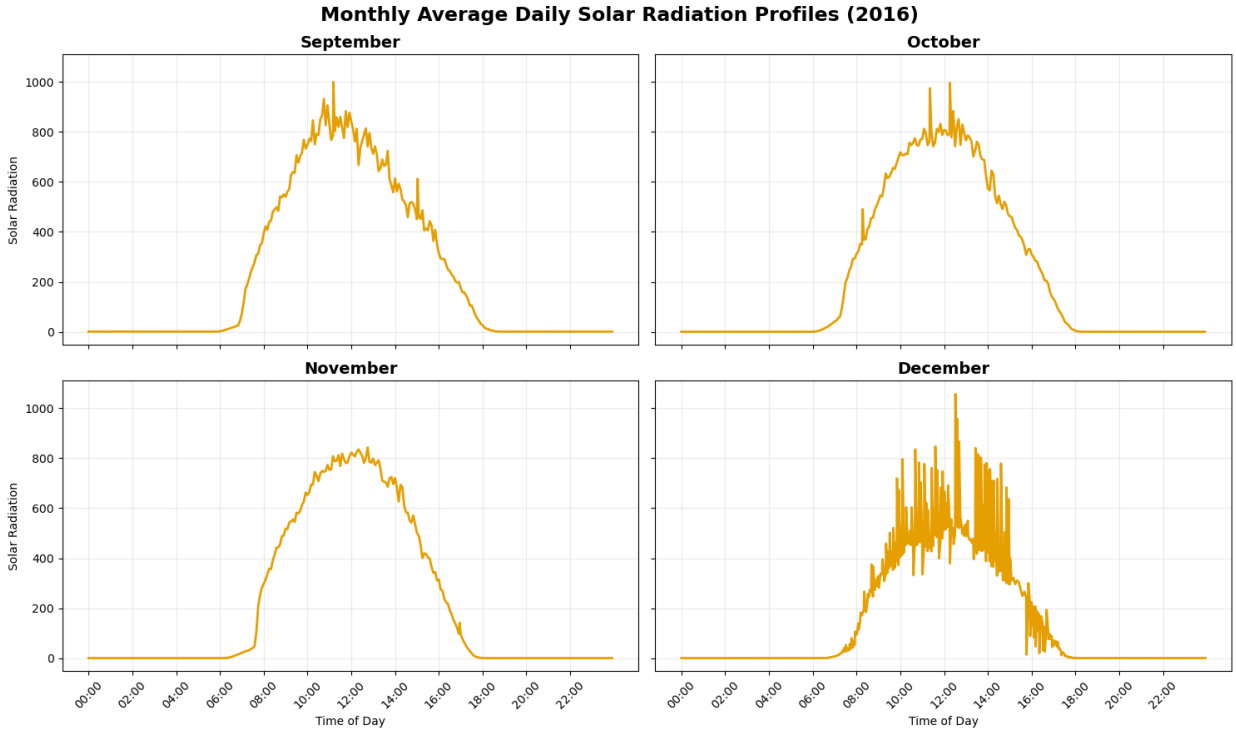


Figure 1. Mean radiation by time of day, per month.

When is radiation highest? Between roughly 10:00 and 14:00, with the clear-sky maximum near 12:00–12:30. That matches solar noon on the peak day ((06:08 + 18:35)/2 ≈ 12:21). Readings are ≈0 before ~06:00 and after ~18:30. The monthly mean curve is lower and flatter than any single clear day because it averages clear and cloudy days: peak mean ≈ 1000 in September, ≈ 800 in November. December is far noisier, which is consistent with very few days being averaged (this should be checked with the per-month day counts that analysis.py prints).

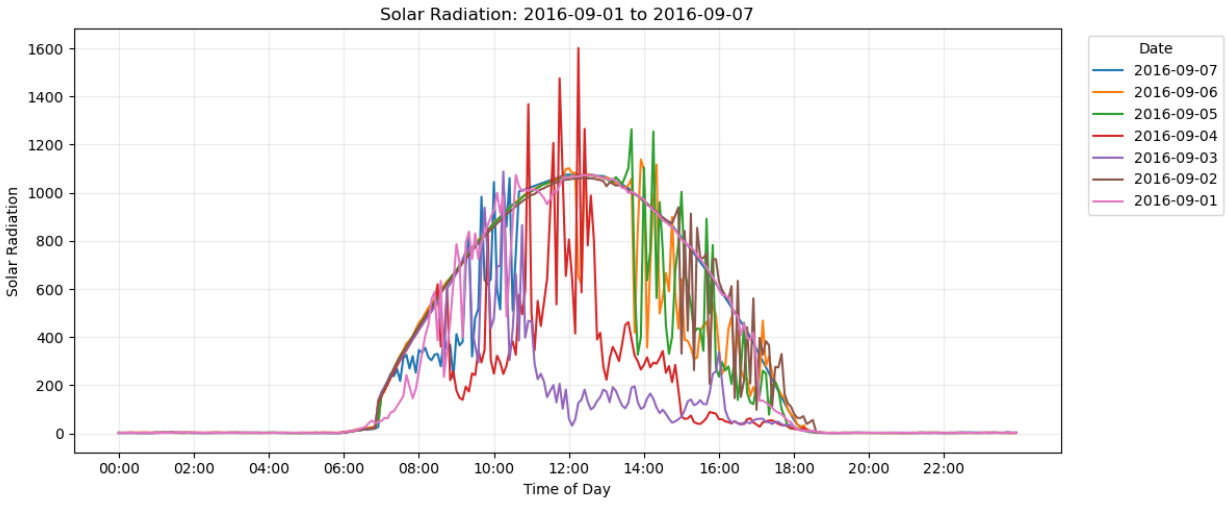


Figure 2. Daily curves, 1–7 Sep. Smooth days trace the clear-sky envelope (~1070); 3 Sep is heavily overcast (~100–200); 4 Sep contains the 1601 spike.

Across the weekly plots, each day is either (i) a smooth clear-sky arch, or (ii) a jagged, depressed curve from passing cloud. Clear-sky days are almost identical to each other within a week, so most of the day-to-day variation in daily mean radiation is cloudiness, not the sun's geometry.

3.2 Radiation versus environmental parameters

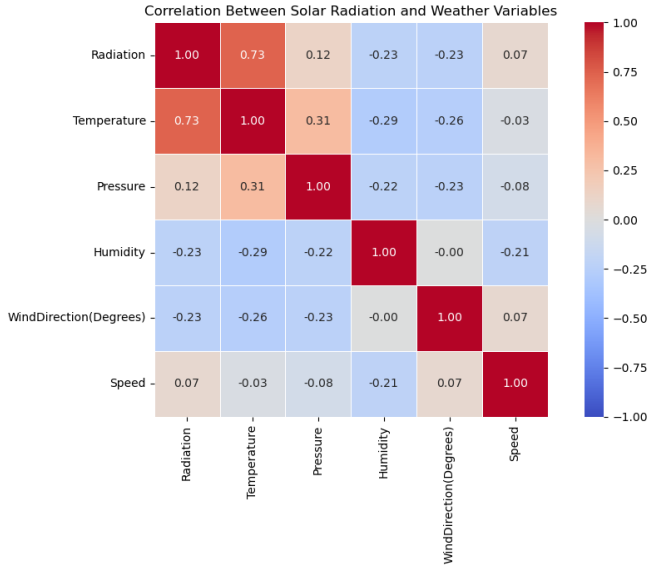


Figure 3. Correlation matrix.

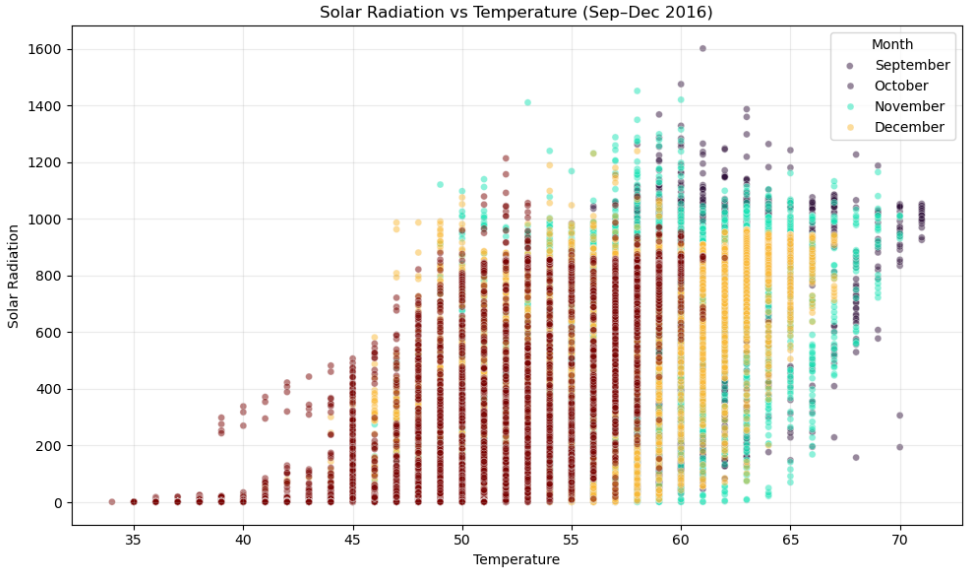


Figure 4. Radiation vs temperature, coloured by month.

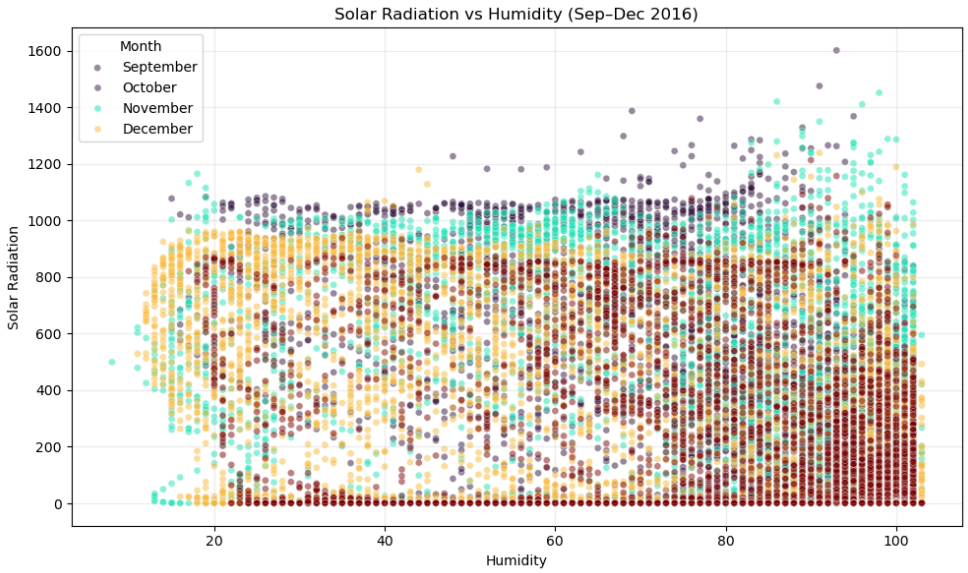


Figure 5. Radiation vs humidity.

Strongest relationship: temperature (r = 0.73). Below ~40 °F radiation is essentially zero; above ~60 °F it can reach 1000+. Temperature forms a ceiling: high radiation only occurs when it is warm, but warm hours can still have low radiation (e.g. late afternoon, or cloud). **Caution:** this correlation is computed over all hours including night, and both variables follow the same daily cycle. Part of the 0.73 is therefore shared time-of-day, not temperature "causing" radiation; the physical direction is radiation heating the air.

Radiation, temperature and humidity. Temperature and humidity are negatively correlated (–0.29); humidity and radiation also (–0.23). Cool hours (night, early morning) are humid and dark; warm midday hours are drier and bright. In Figure 5 the highest readings occur at low-to-moderate humidity, while the dense cluster of low-radiation points sits at 80–100% humidity. High humidity does not *prevent* bright readings (points above 1000 exist near 100%), which is why the correlation is only moderate.

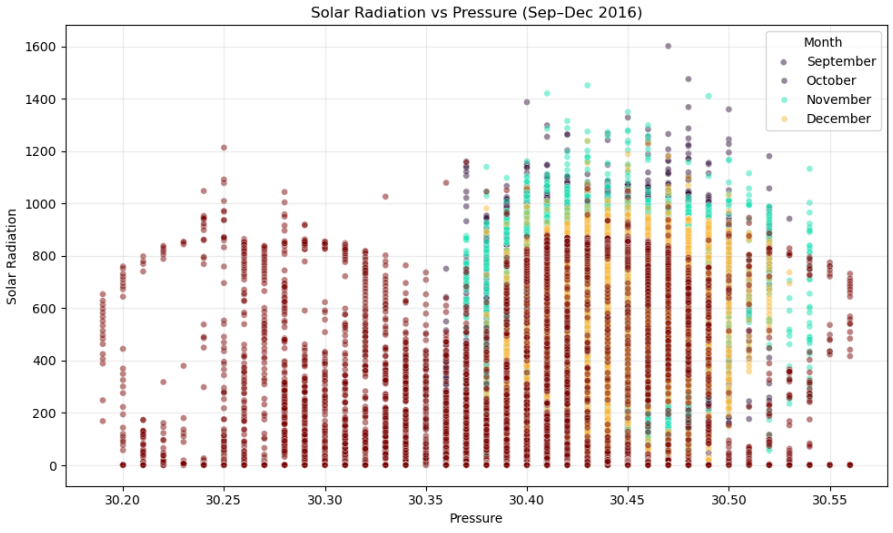


Figure 6. Radiation vs pressure.

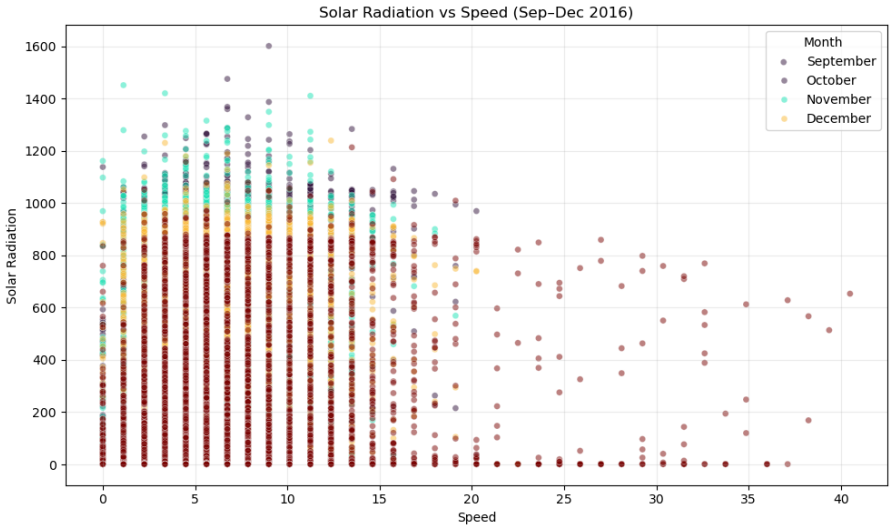


Figure 7. Radiation vs wind speed.

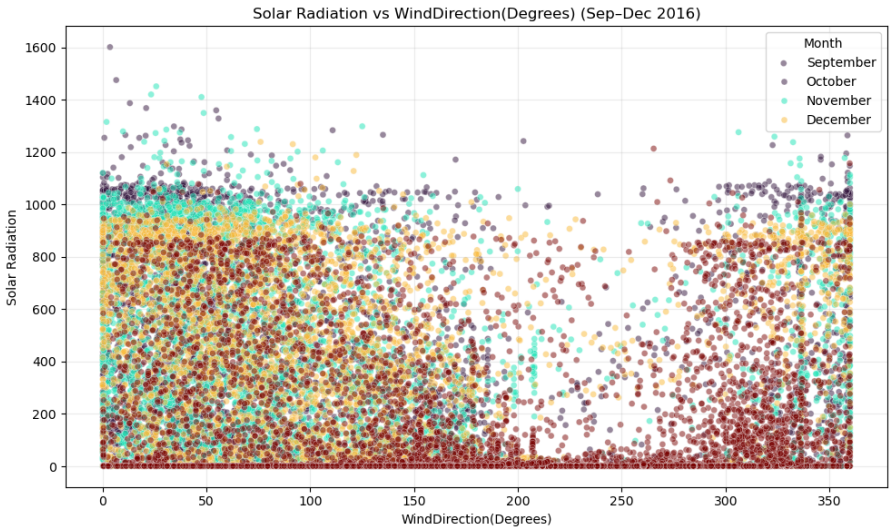


Figure 8. Radiation vs wind direction.

Pressure (r = 0.12) shows little direct link; the lowest pressures (<30.35) occur only in the early part of the record, so pressure partly acts as a proxy for season. **Wind speed** (r = 0.07) has no usable relationship: the rare gusts above ~20 mph all coincide with low-to-moderate radiation, but they are few. **Wind direction** shows a pattern worth noting: very few observations between ~200° and ~280°, and the winds that do occur from 200–300° are mostly low-radiation (night or cloudy) readings. Bright midday readings are concentrated in 0–150° and 300–360°.

3.3 Outliers and unusual observations

- **2016-09-04 spike (1601.26 at 12:15).** Far above every other reading (next highest ≈1450). On a mostly overcast day (Fig. 2, red) this is typical of cloud-edge enhancement, where sunlight reflected from cloud edges adds to the direct beam, or of a sensor glitch. Treat it with caution.
- **Straight-line segments on 8–9 Sep** (Fig. 9) and the truncated trace on 15 Sep (ends ≈17:00) indicate **missing data**; the plot interpolates across the gap. These days must not be trusted for daily means.
- **Overcast days:** 3 Sep, 15 Sep and 21 Oct (morning) stay at 100–300 all day, and 1–6 Oct is persistently cloudy and noisy. 8 Nov shows a sudden drop at ≈12:30.

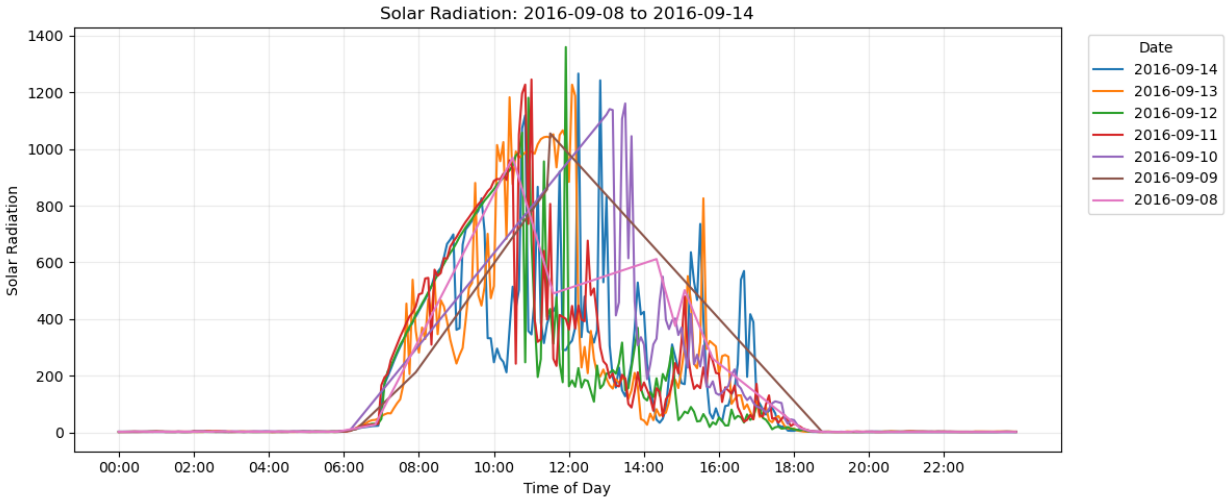


Figure 9. 8–14 Sep: the linear segments on 8 and 9 Sep are interpolated gaps.

3.4 Bonus insights

- **Radiation "switches on" after the listed sunrise.** In late Oct–Nov the curves jump abruptly from ~30 to ~250–300 around 07:30–07:40, although sunrise is ~06:10–06:30 (Fig. 10). The step moves later through the season (≈07:00 in Sep, ≈07:40 in Nov). A plausible explanation is terrain or an obstruction on the eastern horizon that hides the sun for the first ~1–1.5 h, so the effective sunlight window is shorter than sunrise–sunset suggests. This is an inference from the plots; analysis.py produces bonus_sunrise_step.png to quantify it.
- **Seasonal decline in the clear-sky ceiling.** The smooth-day envelope peaks at ≈1070 (Sep), ≈1000–1030 (Oct), ≈950 (early Nov) and ≈900 (mid Nov), while the day length also shrinks. This is the expected seasonal drop in sun elevation, and it explains why the top-5 days all fall in September.

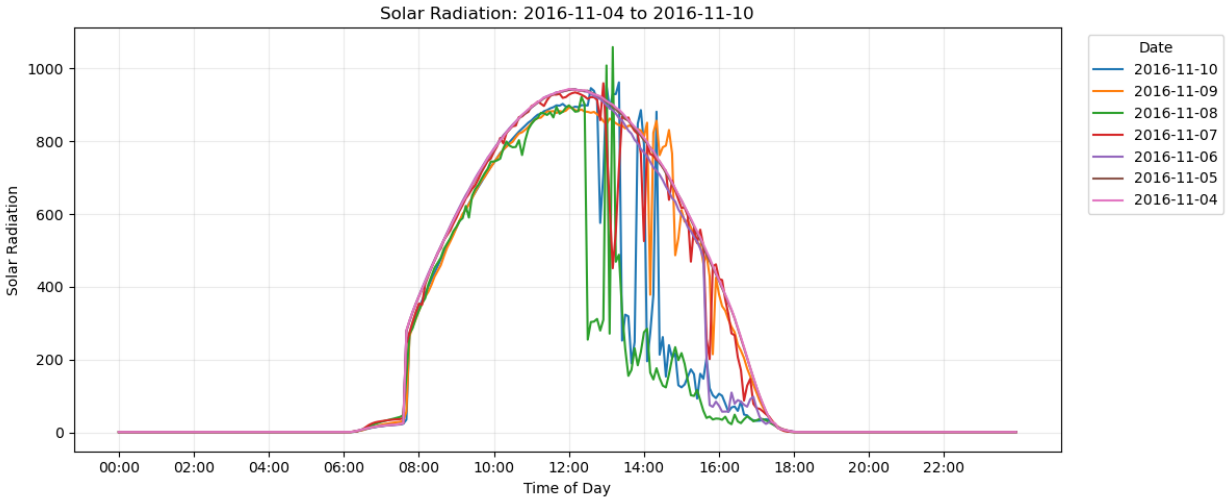


Figure 10. 4–10 Nov: abrupt step at ≈07:40 and a clean clear-sky arch peaking ≈940.

4. Part C — Solar day analysis

Implemented with Pandas groupby/agg and NumPy-based time arithmetic (analysis.py, common.py). The full table is written to outputs/daily_report.csv.

4.1 Peak day

Quantity	Value
Date of highest radiation	2016-09-04
Maximum radiation	1601.26
Time of peak	12:15:04
Temperature	61
Humidity	93
Wind speed	9.0
Pressure / wind direction	30.47 / 3.56°
Sunrise / sunset	06:08:00 / 18:35:00
Peak after sunrise	367 min (6 h 07 min)
Peak before sunset	380 min (6 h 20 min)
Day length / solar noon	12 h 27 min / ≈12:21 (peak ≈6 min before)

The peak is almost exactly at the midpoint of the day, as expected for the sun's highest point. But the 93% humidity and the surrounding overcast curve (Fig. 2) reinforce that the 1601 value is an anomaly rather than a clear-sky maximum.

4.2 Top 5 days by average radiation

Rank	Date	Mean radiation	Max radiation
1	2016-09-19	335.50	1085.75
2	2016-09-02	333.47	1062.01
3	2016-09-22	325.52	1053.17
4	2016-09-25	319.92	1075.29
5	2016-09-28	312.40	1029.30

All five are clear September days (smooth arches in the weekly plots), and their maxima (1030–1090) sit on the clear-sky ceiling, so none is driven by a spike. Note that means include the night-time zeros, so they are lower than daytime averages; the ranking is unaffected. The two days with missing data (8–9 Sep) should be excluded from any ranking if you want to be strict.

5. Bonus — Solar radiation prediction

Implementation: src/model.py. **Target:** Radiation. **Features:** Temperature, Pressure, Humidity, wind speed, wind direction (sin/cos), time of day (sin/cos), day of year, minutes since sunrise, minutes to sunset and the fraction of the daylight window elapsed. Cyclical variables are encoded with sin/cos so that 359° and 1° (or 23:55 and 00:05) are neighbours.

Split: 80/20. Two versions are reported: a random split (as asked), and a **day-grouped split** where whole days are held out. Readings 5 minutes apart are near-duplicates, so the random split leaks information from train to test and over-states accuracy; the grouped split is the honest figure. **Models:** Linear Regression (baseline) and a Random Forest (non-linear, handles the bell-shaped daily curve and interactions). **Metrics:** RMSE and R^2.

5.1 Results

Split	Model	RMSE	R^2
Random 80/20	Linear Regression	__	__
Random 80/20	Random Forest	__	__
Day-grouped 80/20	Linear Regression	__	__
Day-grouped 80/20	Random Forest	__	__

[TO FILL IN] The dataset file was not available when this report was prepared, so no scores are reported here and none have been invented. Run python src/model.py, then copy the values from outputs/model_scores.csv into the table above and the feature ranking from outputs/feature_importance.csv into the paragraph below.

5.2 Which features should matter

Expectation from Part B (to be confirmed by the permutation-importance output): time-of-day / sun-position features (minutes since sunrise, time sin/cos, day fraction) should dominate, because the daily arch is the largest source of variance; temperature should rank next (r = 0.73); humidity should add a little by flagging cloudy, damp conditions; pressure, wind speed and wind direction are expected to contribute little. Expect the Random Forest to beat the linear baseline clearly, since radiation is not linear in time of day.
Cloud-driven spikes are unpredictable from these features, so the residuals will be largest on partly cloudy days.

6. Limitations

- Correlations include night-time rows and shared diurnal cycles; they show association, not causation.
- Units are assumed; days with missing data (8–9, 15–16 Sep) are interpolated in plots and not excluded from daily means.
- Cause of the 1601 peak and of the post-sunrise step are plausible explanations, not proven ones.
- Only four months of one year: seasonal conclusions cannot be generalised, and a model trained on Sep–Dec will not extrapolate to other seasons.

7. Reproducing

See README.md in the repository: place SolarPrediction.csv in data/, run src/analysis.py then src/model.py. The original exploratory notebook is in notebooks/.